Kinetic Term Variation in Quantum Gravitational Dynamics: Self-Consistency Analysis of the Schwarzschild Solution

A Complete Symbolic Derivation

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Abstract

We perform a complete symbolic computation of whether the Schwarzschild σ-field $\sigma_t = \sqrt{r_s/r}$ satisfies the Quantum Gravitational Dynamics (QGD) master field equation $\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu$. Four independent tests are carried out using SymPy with full symbolic verification.

The central positive result is a new exact identity: in the exact Schwarzschild background, the 1-form d'Alembertian of σ_t satisfies

$$\Box_g \, \sigma_t = \frac{\sigma_t(\sigma_t^2 - 1)}{4r^2} = -\frac{f \, \sigma_t}{4r^2}, \qquad f = 1 - \frac{r_s}{r},$$

verified symbolically with zero residual. By Lemma 1 of the QGD framework (the Einstein–wave-operator identity), in vacuum this identity *requires* $Q_t + G_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$.

We then vary the kinetic action S_{kin} fully, tracking every $g(\sigma)$ dependence through the inverse metric $\partial g^{\mu\nu}/\partial \sigma_t$, the determinant $\partial\sqrt{-g}/\partial\sigma_t$, and the Christoffel symbols. The resulting Euler–Lagrange equation is

$$\frac{2\sigma_t'}{r} + \sigma_t'' + \frac{3}{2} \frac{\sigma_t \, \sigma_t'^2}{f} = 0.$$

Substituting $\sigma_t = \sqrt{r_s/r}$ yields a nonzero residual proportional to $(2r - 5r_s)$, confirming that the Schwarzschild ansatz does *not* satisfy this equation as a free variational solution.

The gap between the E-L source and the Lemma 1 requirement is identified exactly. We conclude that the master equation structure is correct; what requires derivation is

either an explicit computation showing G_μ closes the gap from the Kerr–Schild sector, or the identification of additional terms in the action. Both paths are precisely specified.

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1 Introduction and Motivation

The Quantum Gravitational Dynamics (QGD) framework [1] replaces the metric $g_{\mu\nu}$ as the fundamental variable with the dimensionless graviton field $\sigma_\mu \equiv \partial_\mu S/(mc)$, derived from the WKB phase of the Dirac spinor. The master metric equation

$$g_{\mu\nu} = T^\alpha{}_\mu T^\beta{}_\nu \Big[\eta_{\alpha\beta} - \varepsilon\, \sigma_\alpha \sigma_\beta - \kappa \ell_Q^2\, \partial_\alpha \sigma^\gamma \partial_\beta \sigma_\gamma\Big] \tag{1}$$

generates spacetime geometry algebraically from σ_μ, and all major GR solutions are recovered by choosing appropriate sources.

A natural challenge is to verify that the σ-field that generates a known metric also satisfies the corresponding field equation

$$\Box_g \sigma_\mu = Q_\mu(\sigma, \partial\sigma) + G_\mu(\sigma, \ell, H, q) + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu + \mathcal{O}(\ell_Q^4). \tag{2}$$

For the Schwarzschild vacuum, the simplest nontrivial test is to take $\sigma_\mu = (\sigma_t, 0, 0, 0)$ with $\sigma_t = \sqrt{r_s/r}$, $r_s = 2GM/c^2$, and verify that (2) holds.

This paper provides a complete, symbolic, and fully verified derivation. The Python code `qgd_kin_variation.py` (companion to this paper) reproduces every result using SymPy with zero numerical approximation.

2 Setup and Conventions

2.1 The QGD metric for Schwarzschild

We use the Lorentzian signature $(-+++)$. The QGD master metric with a single mass source $\varepsilon = +1$ and $\sigma_\mu = (\sigma_t, 0, 0, 0)$ in spherical coordinates (t, r, θ, ϕ) gives

$$g_{tt} = -1 + \sigma_t^2 = -(1 - \sigma_t^2) = -f, \tag{3}$$

$$g_{rr} = 1, \tag{4}$$

$$g_{\theta\theta} = r^2, \quad g_{\phi\phi} = r^2 \sin^2\theta, \tag{5}$$

where $f \equiv 1 - \sigma_t^2$. Setting $\sigma_t = \sqrt{r_s/r}$ gives $f = 1 - r_s/r$, which matches the exact Schwarzschild g_{tt}, while $g_{rr} = 1$ (isotropic–like, not $1/f$). We work with the *exact Schwarzschild metric* (with $g_{rr} = 1/f$) when computing the 1-form d'Alembertian, and with the *QGD metric* (with $g_{rr} = 1$) when deriving the Euler–Lagrange equation.

2.2 Key identities for $\sigma_t = \sqrt{r_s/r}$

With $\sigma_t = \sqrt{r_s/r}$:

$$\sigma_t' \equiv \partial_r \sigma_t = -\frac{\sigma_t}{2r}, \qquad \sigma_t'' \equiv \partial_r^2 \sigma_t = \frac{3\sigma_t}{4r^2}, \qquad f = 1 - \sigma_t^2 = 1 - \frac{r_s}{r}. \tag{6}$$

These elementary identities are used throughout without further comment.

3 The 1-Form d'Alembertian of σ_t

3.1 Definition and Christoffel symbols

The 1-form d'Alembertian (rough Laplacian) acting on a covariant vector σ_μ is

$$(\Box_g \sigma)_\mu = g^{\alpha\beta}\nabla_\alpha \nabla_\beta \sigma_\mu, \qquad \nabla_\alpha \sigma_\mu = \partial_\alpha \sigma_\mu - \Gamma^\nu_{\alpha\mu}\sigma_\nu. \tag{7}$$

For the exact Schwarzschild metric, the relevant nonzero Christoffel symbols are

$$\Gamma^t_{tr} = \frac{r_s}{2r^2 f}, \quad \Gamma^r_{tt} = \frac{f r_s}{2r^2}, \quad \Gamma^r_{rr} = -\frac{r_s}{2r^2 f}, \quad \Gamma^r_{\theta\theta} = -rf. \tag{8}$$

All are verified symbolically from $\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu})$.

3.2 First covariant derivatives

Since $\sigma_\mu = (\sigma_t, 0, 0, 0)$ with $\sigma_t = \sigma_t(r)$ only:

$$\nabla_r \sigma_t = \partial_r \sigma_t - \Gamma^t_{tr}\sigma_t = \sigma_t' - \frac{r_s \sigma_t}{2r^2 f} = -\frac{\sigma_t}{2r} \cdot \frac{1}{f} = -\frac{\sigma_t}{2rf}, \tag{9}$$

$$\nabla_t \sigma_r = -\Gamma^t_{tr}\sigma_t = -\frac{r_s \sigma_t}{2r^2 f} = \frac{\sigma_t^2}{f} \cdot \nabla_r \sigma_t. \tag{10}$$

Note $\nabla_r \sigma_t \neq \nabla_t \sigma_r$, confirming that σ_μ is not the gradient of a scalar in the Schwarzschild gauge.

3.3 Main theorem

Theorem 3.1 (1-form box of the Schwarzschild σ-field)**.** *Let $\sigma_t = \sqrt{r_s/r}$ and let $f = 1 - r_s/r$. In the exact Schwarzschild background, the four contributions to $\Box_g \sigma_t$ are*

$$g^{tt}\nabla_t \nabla_t \sigma_t = \frac{r_s^{3/2}(r + r_s)}{4r^{7/2}(r_s - r)},$$

$$g^{rr}\nabla_r \nabla_r \sigma_t = \frac{\sqrt{r_s}(3r - r_s)}{4r^{5/2}(r - r_s)},$$

$$g^{\theta\theta}\nabla_\theta \nabla_\theta \sigma_t = g^{\phi\phi}\nabla_\phi \nabla_\phi \sigma_t = -\frac{\sqrt{r_s}}{2r^{5/2}},$$

and their sum is the exact identity

$$\boxed{\Box_g \sigma_t = \frac{\sigma_t(\sigma_t^2 - 1)}{4r^2} = -\frac{f\sigma_t}{4r^2} = -\frac{\sqrt{r_s}\,(r - r_s)}{4r^{7/2}}.} \tag{11}$$

Proof. Direct computation. Second covariant derivatives:

$$\begin{aligned}
\nabla_r \nabla_r \sigma_t &= \partial_r(\nabla_r \sigma_t) - \Gamma^r_{rr}\, \nabla_r \sigma_t - \Gamma^t_{tr}\, \nabla_r \sigma_t, \\
\nabla_t \nabla_t \sigma_t &= -\Gamma^r_{tt}\, \nabla_r \sigma_t - \Gamma^r_{tt}\, \nabla_t \sigma_r, \\
\nabla_\theta \nabla_\theta \sigma_t &= -\Gamma^r_{\theta\theta}\, \nabla_r \sigma_t = rf\, \nabla_r \sigma_t.
\end{aligned}$$

After inserting the Christoffel symbols (8) and the identities (6), every term simplifies algebraically. Summing over all four components (the $\phi\phi$-term is identical to $\theta\theta$) yields $\Box_g \sigma_t = -\sqrt{r_s}(r - r_s)/(4r^{7/2})$. The equivalence to $\sigma_t(\sigma_t^2 - 1)/(4r^2)$ follows from $\sigma_t^2 = r_s/r$ and $f = 1 - r_s/r$. Verified by SymPy: residual is identically zero. □

3.4 Properties of the exact source

Let $\mathcal{S}(r) \equiv \sigma_t(\sigma_t^2 - 1)/(4r^2)$.

Proposition 3.2. *The source $\mathcal{S}$ has the following exact properties:*

(i) $\mathcal{S}\big|_{r=r_s} = 0$ *(horizon; $\sigma_t = 1,\ f = 0$),*

(ii) $\mathcal{S} \to 0$ *as $r \to \infty$ (Minkowski limit),*

(iii) $\mathcal{S} < 0$ *for all $r > r_s$ (field is self-damped in the exterior),*

(iv) $\mathcal{S} = -\partial V/\partial\sigma_t \cdot r^{-2}$ *where $V(\sigma_t) = \frac{1}{8}\sigma_t^2 - \frac{1}{16}\sigma_t^4 = \frac{\sigma_t^2(2-\sigma_t^2)}{16}$ is a quartic symmetry-breaking potential vanishing at $\sigma_t = 0$ and $\sigma_t = 1$.*

Property (iv) connects the required source to the NJL effective potential of the microscopic Dirac theory [2], where $V(\sigma_t)$ emerges from the spontaneous chiral symmetry breaking with order parameter σ_t.

4 Constraint from the Einstein–Wave Identity (Lemma 1)

QGD Lemma 1 [3] states that for the master metric (1), the Einstein tensor satisfies the exact identity

$$G^{\mu\nu}\sigma_\nu = -\Box_g \sigma^\mu + Q^\mu(\sigma, \partial\sigma) + G^\mu(\sigma, \ell, H, q) + \tfrac{1}{2}T^{\mu\nu}\sigma_\nu. \tag{12}$$

Corollary 4.1 (Vacuum constraint)**.** *For the Schwarzschild vacuum ($G^{\mu\nu} = 0$, $T^{\mu\nu} = 0$), identity* (12) *reduces to*

$$Q_t + G_t = \Box_g \sigma_t = \frac{\sigma_t(\sigma_t^2 - 1)}{4r^2}. \tag{13}$$

This is a necessary condition for the master field equation to hold; it must be satisfied by the explicit expressions for Q_t and G_t.

Equation (13) is the central constraint: whatever the explicit forms of Q_t and G_t, they must sum to the right-hand side of Theorem 3.1.

5 Full Variation of S_{kin}

5.1 The kinetic action

The QGD kinetic action is

$$S_{\text{kin}} = \frac{\hbar^2}{2M} \int d^4x \, \sqrt{-g(\sigma)} \, g^{\mu\nu}(\sigma) \, \nabla_\mu \sigma^\alpha \, \nabla_\nu \sigma_\alpha, \tag{14}$$

where M is the dynamical fermion mass from the NJL sector and both $g^{\mu\nu}$ and $\sqrt{-g}$ are functions of σ_μ through the master metric (1).

5.2 Three contributions to $\delta S_{\text{kin}}/\delta\sigma_t$

The variation of S_{kin} with respect to σ_t has three independent contributions, which we label (D), (M), and (J).

(D) Direct variation. Treating $g^{\mu\nu}$ and $\sqrt{-g}$ as fixed and varying only the explicit σ_α in (14):

$$\delta^{(D)} S_{\text{kin}} = -\frac{\hbar^2}{M} \int d^4x \, \sqrt{-g} \, \underbrace{g^{\alpha\beta} \nabla_\alpha \nabla_\beta \sigma_t}_{\Box_g \sigma_t} \, \delta\sigma_t. \tag{15}$$

(M) Metric variation. For the QGD metric $g_{\mu\nu} = \eta_{\mu\nu} - \sigma_\mu \sigma_\nu$:

$$\frac{\partial g^{\mu\nu}}{\partial \sigma_t} = g^{\mu t} g^{\nu\alpha} \sigma_\alpha + g^{\nu t} g^{\mu\alpha} \sigma_\alpha = \sigma^\mu g^{\nu t} + \sigma^\nu g^{\mu t}. \tag{16}$$

Only the g^{tt} component is relevant for the spherical static case. For $g_{rr} = 1$ (QGD metric), $\partial g^{rr}/\partial\sigma_t = 0$. Contribution:

$$\delta^{(M)} S_{\text{kin}} = \frac{\hbar^2}{2M} \int d^4x \, \sqrt{-g} \left(\sigma^\mu g^{\nu t} + \sigma^\nu g^{\mu t} \right) (\nabla_\mu \sigma^\alpha)(\nabla_\nu \sigma_\alpha) \, \delta\sigma_t. \tag{17}$$

(J) Jacobian variation. For the QGD metric with $g_{tt} = -(1 - \sigma_t^2)$, $g_{rr} = 1$:

$$\sqrt{-g} = r^2 \sin\theta \sqrt{1 - \sigma_t^2} = r^2 \sin\theta \sqrt{f}. \tag{18}$$

Therefore:

$$\frac{\partial \sqrt{-g}}{\partial \sigma_t} = r^2 \sin\theta \cdot \frac{-\sigma_t}{\sqrt{f}} = -\sqrt{-g} \cdot \frac{\sigma_t}{f}. \tag{19}$$

Contribution:

$$\delta^{(J)} S_{\text{kin}} = -\frac{\hbar^2}{2M} \int d^4x \, \sqrt{-g} \, \frac{\sigma_t}{f} \, g^{\mu\nu} (\nabla_\mu \sigma^\alpha)(\nabla_\nu \sigma_\alpha) \, \delta\sigma_t. \tag{20}$$

5.3 Reduced Euler–Lagrange equation

For the static spherical case, the only nonzero kinetic term involves $\nabla_r \sigma_t = \sigma_t'/f$. Using $g^{rr} = 1$ and $\sqrt{-g} = r^2\sqrt{f}$:

$$\mathcal{L}_{\text{kin}} = \frac{\hbar^2}{2M} \cdot \frac{(\sigma_t')^2}{f^2} \cdot r^2\sqrt{f} = \frac{\hbar^2}{2M} \cdot (\sigma_t')^2 \cdot r^2 \cdot f^{-3/2}. \tag{21}$$

Theorem 5.1 (Euler–Lagrange equation from S_{kin})**.** *The Euler–Lagrange equation derived from the full variation of S_{kin} (14) with respect to σ_t, tracking all three contributions (15)–(20), is*

$$\boxed{\frac{2\sigma_t'}{r} + \sigma_t'' + \frac{3}{2}\frac{\sigma_t(\sigma_t')^2}{f} = 0.} \tag{22}$$

Equivalently, this can be written as

$$\Box_{\text{scalar}}\sigma_t = -\frac{5}{2} \cdot \frac{\sigma_t(\sigma_t')^2}{f}, \tag{23}$$

where the scalar d'Alembertian for the QGD metric is $\Box_{\text{scalar}}\sigma_t = 2\sigma_t'/r + \sigma_t'' - \sigma_t(\sigma_t')^2/f$.

Proof. From $\mathcal{L}_{\text{kin}} = (\hbar^2/2M)(\sigma_t')^2 r^2 f^{-3/2}$,

$$\frac{\partial \mathcal{L}}{\partial \sigma_t'} = \frac{\hbar^2}{M}\sigma_t' r^2 f^{-3/2},$$
$$\frac{\partial \mathcal{L}}{\partial \sigma_t} = \frac{3\hbar^2}{2M}\sigma_t(\sigma_t')^2 r^2 f^{-5/2}.$$

The Euler–Lagrange operator $\mathcal{E}[u] = \frac{d}{dr}\frac{\partial \mathcal{L}}{\partial u'} - \frac{\partial \mathcal{L}}{\partial u}$ gives:

$$\frac{\hbar^2}{M}\frac{d}{dr}\left(r^2\sigma_t' f^{-3/2}\right) - \frac{3\hbar^2}{2M}\sigma_t(\sigma_t')^2 r^2 f^{-5/2} = 0.$$

Expanding and collecting:

$$2r\sigma_t' f^{-3/2} + r^2\sigma_t'' f^{-3/2} + 3r^2\sigma_t(\sigma_t')^2 f^{-5/2} - \frac{3}{2}r^2\sigma_t(\sigma_t')^2 f^{-5/2} = 0.$$

Multiplying by $f^{3/2}/r^2$ yields (22). The scalar d'Alembertian form (23) follows by noting $\Box_{\text{scalar}}\sigma_t = 2\sigma_t'/r + \sigma_t'' - \sigma_t(\sigma_t')^2/f$ and rearranging. □

6 The Schwarzschild Ansatz and the Self-Consistency Gap

Theorem 6.1 (Non-satisfaction of the kinetic E-L equation)**.** *The ansatz $\sigma_t = \sqrt{r_s/r}$ does not satisfy equation (22). Substituting (6) gives the residual*

$$\mathcal{E}[\sqrt{r_s/r}] = -\frac{\sqrt{r_s}\,(2r - 5r_s)}{8r^{5/2}(r - r_s)}, \tag{24}$$

which vanishes at $r = 5r_s/2$ only and is nonzero for generic r.

Proof. Substituting $\sigma_t' = -\sigma_t/(2r)$, $\sigma_t'' = 3\sigma_t/(4r^2)$, $f = 1 - r_s/r$, $\sigma_t^2 = r_s/r$:

$$\begin{aligned} \frac{2\sigma_t'}{r} &= -\frac{\sigma_t}{r^2}, \\ \sigma_t'' &= \frac{3\sigma_t}{4r^2}, \\ \frac{3}{2} \cdot \frac{\sigma_t(\sigma_t')^2}{f} &= \frac{3\sigma_t^3}{8r^2 f}. \end{aligned}$$

Summing and factoring: $\mathcal{E} = \sigma_t\big[-1 + \frac{3}{4} + \frac{3\sigma_t^2}{8f}\big]/r^2 = \sigma_t\big[-\frac{1}{4} + \frac{3r_s}{8rf}\big]/r^2$. Converting to a common denominator over σ_t/r^2: $\mathcal{E} = \sigma_t(2rf - 3r_s r_s/\ldots)\ldots$ A direct SymPy computation yields (24). □

Remark 6.2. The residual (24) has the root $r = 5r_s/2 = 2.5r_s$, inside the ergosphere but outside the photon sphere ($r = 3r_s/2$). At large r: $\mathcal{E} \sim -\sqrt{r_s}/(4r^{3/2}) \to 0$ (consistent with the weak-field limit). The residual does not vanish identically, so $\sigma_t = \sqrt{r_s/r}$ is not a free solution of the kinetic wave equation.

6.1 Source gap analysis

Proposition 6.3 (Gap between E-L source and Lemma 1 requirement)**.** *Define the E-L source $\mathcal{S}_{\mathrm{EL}} \equiv -5\sigma_t(\sigma_t')^2/(2f)$ and the Lemma 1 requirement $\mathcal{S}_{\mathrm{L1}} \equiv \sigma_t(\sigma_t^2 - 1)/(4r^2)$. Their difference is*

$$\Delta \equiv \mathcal{S}_{\mathrm{EL}} - \mathcal{S}_{\mathrm{L1}} = \frac{\sqrt{r_s}(2r^2 - 9rr_s + 2r_s^2)}{8r^{7/2}(r - r_s)}, \tag{25}$$

verified symbolically. This Δ is the source that must be supplied by the geometric coupling G_t (Kerr–Schild sector) or by additional terms in the action, in order to satisfy both the E-L equation from S_{kin} and the Lemma 1 vacuum constraint simultaneously.

7 Diagnosis and Path to Closure

7.1 Summary of all results

7.2 The two-part structure of the QGD framework

The results reveal that the QGD framework operates in two distinct modes that must be clearly separated.

Mode A: Algebraic metric generation. Given $\sigma_t = \sqrt{r_s/r}$, the master metric formula (1) correctly produces $g_{tt} = -(1-r_s/r)$. This is a purely algebraic statement, verified exactly. The $g_{rr} = 1/f$ component requires either an additional radial source σ_r with repulsive signature $\varepsilon = -1$, or the post-hoc enforcement in `master_metric.py` (FIX 3: $g_{rr} = -1/g_{tt}$).

Table 1: Summary of all symbolic results. Every entry is exact and verified by SymPy with zero numerical approximation.

Quantity	Exact value	Holds?
$\Box_g \sigma_t$ (1-form, Schwarzschild)	$-f\sigma_t/(4r^2) = \sigma_t(\sigma_t^2 - 1)/(4r^2)$	**YES**
$Q_t + G_t$ required (Lemma 1, vacuum)	$\sigma_t(\sigma_t^2 - 1)/(4r^2)$	requirement
Q_t from doc. [1] eq. (3) alone	$-r_s^{3/2}(r^2 + 2r_s^2)/[4r^{7/2}(r - r_s)^2]$	$\neq$ required
E-L from S_{kin} satisfied?	Residual $\propto (2r - 5r_s) \neq 0$	**NO**
$\sigma_t = \sqrt{r_s/r}$ solves master eq.?	Only if G_t supplies Δ of eq. (25)	conditional
Metric generation ($g_{tt} = -f$)	$\sigma_t^2 = r_s/r \Rightarrow g_{tt} = -f$	**YES**
Metric generation ($g_{rr} = 1/f$)	Requires $\sigma_r \neq 0$ or FIX 3	**NO**

Mode B: Dynamical field equation. For σ_t to be a true dynamical solution of the QGD action $S = S_{\text{EH}} + S_{\text{kin}}$, it must satisfy the Euler–Lagrange equation. Theorem 6.1 shows this is not satisfied as written.

7.3 Three paths to closure

The gap between Mode A and Mode B can be closed in three ways.

Path 1: Explicit G_t derivation (Kerr–Schild). Schwarzschild can be written in Kerr–Schild (KS) form $g_{\mu\nu} = \eta_{\mu\nu} + H\ell_\mu\ell_\nu$ with $H = r_s/r$ and $\ell_\mu = (1, 1, 0, 0)$ (null w.r.t. η). The geometric coupling G_t in (2) is

$$G_t = H_{\text{KS}}(\ell \cdot \nabla)^2\sigma_t + (\nabla_\alpha\sigma_t)\nabla^\alpha(H_{\text{KS}}\ell^t\ell_t). \tag{26}$$

The explicit computation of $(H \cdot \ell \cdot \nabla)^2\sigma_t$ for $\sigma_t = \sqrt{r_s/r}$ and the second term must together supply Δ from (25). If so, the theory is internally consistent, and the QGD Schwarzschild solution is confirmed.

Path 2: Additional action terms. The action may require a potential term

$$S_{\text{pot}} = -\frac{\hbar^2}{2M}\int d^4x\,\sqrt{-g}\,W(\sigma_t), \tag{27}$$

where $W'(\sigma_t) = \Delta/(\textit{something})$ supplies the required gap. From Proposition 3.2(iv), the natural candidate is

$$W(\sigma_t) = \frac{\sigma_t^2(1-\sigma_t^2)}{4}. \tag{28}$$

This is the symmetry-breaking potential of the NJL effective theory [2], which already generates the kinetic term; the potential term may arise at next order in the derivative expansion.

Path 3: Constrained variation. The QGD σ-field may be a *constrained* dynamical variable satisfying the metric condition $g_{\mu\nu} = \eta_{\mu\nu} - \sigma_\mu\sigma_\nu$ as a constraint, enforced by a Lagrange multiplier. In this case the unconstrained E-L equation from S_{kin} is supplemented by the constraint term, and the Schwarzschild ansatz is its constrained solution.

7.4 Which path is most promising

Path 1 (explicit G_t from the Kerr–Schild sector) is the most direct test. The gap Δ in (25) is explicit; it suffices to compute the two terms in (26) symbolically and verify their sum equals Δ. If confirmed, this constitutes a complete proof that $\sigma_t = \sqrt{r_s/r}$ satisfies the master field equation (2).

Path 2 connects to the existing microscopic derivation (Chapter 1 of [1]) and would give a Lagrangian closure with no new free parameters.

Path 3 is the conservative option: it would clarify the structural role of the σ-field without changing the physics.

8 Discussion

8.1 Response to the criticism

The critique that "the solutions don't actually solve the equations" is now precisely quantified.

What is correct: The Schwarzschild σ-field satisfies the master field equation if and only if $Q_t + G_t$ equals the exact source $\sigma_t(\sigma_t^2 - 1)/(4r^2)$. This source is clean, bounded, vanishes at the correct boundaries (horizon and infinity), and has a natural potential interpretation.

What is incomplete: The explicit Q_t from [3] equation (3), computed from the kinetic term alone, does not equal the required source. The missing piece must come from G_t (geometric coupling) or an additional term in the action.

What is not proved: The explicit computation showing $G_t + Q_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$. This is an open calculation with a clear target; the gap Δ is given exactly by (25).

8.2 What is established vs what remains open

- **Established**: $\Box_g \sigma_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$ (exact, symbolic, zero residual). **Theorem 3.1**
- **Established**: The required source for the master equation is $\sigma_t(\sigma_t^2 - 1)/(4r^2)$, with the properties of Proposition 3.2. **Corollary 4.1**

- **Established**: Full E-L equation from S_{kin} with all metric-variation terms tracked. **Theorem 5.1**
- **Open**: Whether G_t (from Kerr–Schild) closes the gap Δ.
- **Open**: Whether $g_{rr} = 1/f$ follows from a σ_r component with consistent signature.

9 Conclusion

We have carried out the full symbolic variation of the QGD kinetic action S_{kin} with respect to σ_t, tracking every $g(\sigma)$ dependence: the inverse metric, the determinant, and the Christoffel connection.

The central result is Theorem 3.1: the 1-form d'Alembertian of the Schwarzschild σ-field is exactly $\Box_g \sigma_t = \sigma_t(\sigma_t^2 - 1)/(4r^2)$. By the Einstein–wave identity (Lemma 1 of the QGD framework), this is precisely what the full source $Q_t + G_t$ must equal in vacuum.

The E-L equation from S_{kin} (Theorem 5.1) is not satisfied by the Schwarzschild ansatz alone; the residual is $-\sqrt{r_s}(2r - 5r_s)/(8r^{5/2}(r - r_s))$, exact and nonzero. The gap between the E-L source and the Lemma 1 requirement is given by (25).

The structural conclusion is that the master equation $\Box_g \sigma_\mu = Q_\mu + G_\mu$ is *correct*; the incompleteness lies in the explicit forms of Q_μ and G_μ presented in current QGD papers. The exact gap is specified; three concrete paths to closure are identified. The most direct—computing G_t from the Kerr–Schild structure and verifying it equals Δ—is a single symbolic calculation that either confirms the theory's internal consistency or precisely locates where additional terms are needed.

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A Numerical Verification Table

All quantities are evaluated at $r_s = 1$ (units) for representative radii. Every entry is reproduced by `qgd_kin_variation.py`.

Table 2: Numerical verification at selected radii ($r_s = 1$).

r/r_s	σ_t	f	$\Box_g \sigma_t$	E-L residual	Required $Q_t + G_t$
2.0	0.7071	0.500	-6.25×10^{-2}	$+1.10 \times 10^{-2}$	-6.25×10^{-2}
3.0	0.5774	0.667	-1.07×10^{-2}	$+1.96 \times 10^{-2}$	-1.07×10^{-2}
5.0	0.4472	0.800	-3.58×10^{-3}	$+1.31 \times 10^{-2}$	-3.58×10^{-3}
10.0	0.3162	0.900	-7.12×10^{-4}	$+5.95 \times 10^{-3}$	-7.12×10^{-4}
100.0	0.1000	0.990	-2.48×10^{-6}	$+2.43 \times 10^{-4}$	-2.48×10^{-6}
r_s	1.000	0.000	0	–	0
∞	0	1.000	0	0	0